OPERATIONS RESEARCH & DECISION SCIENCE · GHENT, BELGIUM

Sajjad *(Sam)* Hedayati

I build the model, *then the tool people run it in.*

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PROFILE

Operations researcher with **around eight years of professional experience**. I take operational problems that no off-the-shelf tool covers — capacity, inspection, scheduling and routing under uncertainty — formulate them, build the algorithm that makes them solvable at realistic size, and deliver the result as an application the people who own the decision can run themselves. Since 2022 I have done this as a project researcher with Ghent University and Flanders Make on an industrial partner's own data: the simulation and analytics tool I built there provided **the key evidence behind Flanders Make's decision to continue funding the programme**. Before that, three years in planning, industrial engineering and process consulting, presenting to boards and management teams.

Research focus: optimization under uncertainty for production and logistics planning — two-stage stochastic programming, exact and decomposition methods, and learning from data with provable guarantees. Three journal articles, including *Computers & Operations Research* (2026), and eight peer-reviewed conference contributions; reviewer for three international OR journals.

EXPERIENCE

Jan 2022 – present

Project Researcher

Ghent University & Flanders Make · Dept. of Industrial Systems Engineering and Product Design · REMASC programme since Oct 2024

- **They had no tool for it; I built one, and it changed a funding decision.** A discrete-event simulation and analytics application — a digital twin of the partner's cellular remanufacturing floor: stochastic failures, rework loops, WIP and spare-parts inventory, replenishment policies, resource utilisation, queues, operator-skill effects and cost — with automated scenario comparison of dispatching rules, staffing and shift schedules. Delivered as a self-service application so the partner's engineers could test changes themselves; it provided the key evidence behind Flanders Make's decision to continue funding the programme.
- **Built the shift-scheduling engine for the partner's operators.** A simulation-optimization method of my own, solved with Gurobi. Operator skills are modelled per job type, operator, cell and machine, and each arrival carries the partner's own classification label, which sets its route. Handed over with a technical booklet on the model and the code, and a user guide with a narrated video. It runs from a browser app or a JSON input. The REMASC partners own it.
- **Solved what the solver could not.** Formulated inspection-station and capacity allocation as integer programmes, then built two Lagrangian relaxation approaches with greedy repair and hill-climbing search that return near-optimal plans with bounded gaps where the commercial solver stalls at the time limit; on the largest benchmark, runtime fell from 418.50 s to 23.31 s with the reported gap below 8%. Benchmarked against the exact solver and a problem-specific genetic algorithm; published in *Computers & Operations Research* (2026).
- **Made an intractable case exact.** Derived an exact dynamic program for the uncapacitated case: identical optima on all 10,000 tested instances and an 8.6× average speedup at 100 stages, verified against an independent branch-and-bound implementation.
- **Showed what modelling uncertainty is worth.** On an independent holdout sample, the stochastic inspection plan cost 9.3% less than expected-value planning; a certified scenario reduction then cut 1,000 scenarios to 50 without giving back the tested out-of-sample result.
- **Planned under uncertain supply.** Multi-period disassembly planning and scheduling with uncertain return volumes and material condition: two-stage stochastic programming, sample average approximation with statistical gap bounds and scenario reduction, benchmarked against a deterministic model to quantify what the stochastic plan is actually worth.
- **Priced a model's accuracy.** Modelled how an imperfect condition-triage classifier propagates error into downstream throughput and cost, turning "is this model good enough?" into a figure a business can decide on.
- **Made a learned optimizer replace a hand-tuned one.** Trained a PPO agent to learn Lagrangian multiplier updates in place of hand-tuned subgradient schedules.

Feb 2022 – present

Teaching Assistant

Ghent University · Faculty of Engineering and Architecture

Run the exercise sessions for the full course *Probability, Quality Engineering and Industrial Statistics* and deliver theory lectures, hands-on in R: exploratory data analysis, hypothesis testing, regression, ANOVA, design of experiments, acceptance sampling, statistical process control, and process and measurement-system capability.

- **Co-supervisor, M.Sc. thesis (2025–2026):** *A Learning-Based Approach to Lagrangian Relaxation in the Generalized Assignment Problem* — a graph neural network predicts the subgradient search direction and a reinforcement-learning agent corrects the Polyak step size, trained on small instances and transferred unchanged to larger ones; reduced multiplier oscillation and gave better final bounds.

Sep – Dec 2021

Visiting Researcher

Eindhoven University of Technology (TU/e) · OPAC group, host Prof. dr. Tom Van Woensel

Routing in city logistics and warehousing: joint order batching and picker routing, last-mile delivery with alternative locations, and mobile parcel lockers. Coursework with BETA, the Dutch research school for Operations Management & Logistics. Led to two joint journal articles and a joint presentation at the 9th INFORMS TSL Workshop.

Dec 2015 – Jul 2021 · Tehran, Iran · six positions, two of them concurrent, totalling three years and three months, in planning, industrial engineering and business-process analysis. Common thread: BPMN process modelling, SQL extraction and merging against production databases, R for analytics, and process optimization delivered in operating companies — from analysis through to implemented change.

May – Jul 2021

Business Analyst
APSAN Co

Process modelling and optimization, workflow and resource allocation, KPI definition and tracking; reported to the board.

Dec 2020 – Feb 2021

Business Analyst / Process Consultant
Baroline

Feasibility studies, stakeholder workshops and process-reengineering proposals, followed through with technical teams.

Jun – Nov 2019

Industrial Engineer
Hacoupian · garment manufacturer

Bottleneck identification on a live garment production floor and process optimization at head office; analysis reported to the board.

Apr 2018 – Jun 2019

Planning Specialist
SmartComm · start-up

Owned project planning, timetabling and scheduling across competing teams; business-model revision and budget management.

Dec 2015 – Nov 2016

Business Process Analyst
SarvRayaneh

Process modelling and improvement; contributed to building an internal workflow platform and designed the data relationships behind its reporting.

Feb – Sep 2016

Business Process Analyst
Entekhab Group · petrochemical facility · alongside SarvRayaneh

Process designer on a SAP Materials Management implementation: blueprinted the procurement, material-flow and stock-movement processes, settled with the business before they were built into the system. Process design, not configuration.

RESEARCH CONTRIBUTIONS

Scenario reduction

Stochastic programming and certified scenario reduction
- Two-stage stochastic integer programs for inspection allocation, solved by sample average approximation and L-shaped / Benders decomposition with statistical optimality-gap estimation.
- A scenario-reduction certificate: a computable bound on the error of a reduced scenario set that holds uniformly over all first-stage decisions, with stratified sampling, robust variants and scalability studies.
- A contextual extension in which scenario reduction and plans depend on observed covariates.

Current line

Learning decision-dependent feasible sets
- Two-stage programs whose recourse domain depends on the first-stage decision and on an unknown contextual parameter: proved existence, lower semicontinuity and stability, and showed that a single-sided indicator formulation is exact for threshold activation.
- Estimated the domain with honest random forests and nearest neighbours, with finite-sample optimality bounds, activation recovery under a margin condition, and a calibration that certifies an inner domain; synthetic study over 720 data sets.

Exact methods

Allocation and scheduling
- Integer programming formulations for inspection-station allocation with delayed inspection, strengthened by valid inequalities, lazy cuts and warm starts; two Lagrangian relaxations with dual-guided repair and local search.
- An exact dynamic-programming algorithm for the uncapacitated variant, with correctness proof.
- Stochastic lot-sizing and scheduling for quality-differentiated disassembly (logic-based Benders).

Logistics

Logistics and warehousing · with TU/e and Cardiff University

- Joint order batching and picker routing with alternative pick locations, modelled as a clustered generalized vehicle routing problem; two MILP formulations and a cost-matrix rule for rectangular warehouses; instances up to 180 nodes.
- Re-supplying autonomous mobile parcel lockers: meeting-point selection and truck routing with multiple time windows and time-dependent parking costs, solved by cluster-based simulated annealing on a new benchmark set generated from Solomon data and validated against CPLEX.

EDUCATION

Jan 2022 – Dec 2026

Ph.D. Industrial Engineering

Ghent University · Faculty of Engineering and Architecture

Thesis: *Stochastic Optimization Models for Allocation of Quality Inspection in Flexible Assembly*, within the Flanders Make REMASC programme on remanufacturing. Supervisors: Prof. dr. ir. Stijn De Vuyst and Prof. dr. Birger Raa. Internal defence December 2026; public defence expected January/February 2027.

Sep 2018 – Dec 2021

Doctoral studies, Industrial Engineering

K. N. Toosi University of Technology · Tehran

Program not completed. Thesis project: *Vehicle Routing Problem with Non-Stationary Demand Points*, under Prof. dr. Mostafa Setak. The routing and warehousing work of this period was completed and published with Prof. Tom Van Woensel (TU/e) and Prof. Emrah Demir (Cardiff University).

Sep 2014 – Feb 2017

M.Sc. Industrial Engineering

University of Tehran

Sep 2009 – Sep 2013

B.Sc. Industrial Engineering

Shahid Beheshti University of Medical Sciences · Tehran

PUBLICATIONS

Journal articles

J1 Hedayati, S., De Vuyst, S., Raa, B. (2026). Inspection station allocation in a production line with delayed inspection. *Computers & Operations Research*, 194, 107588.
doi:10.1016/j.cor.2026.107588 · Web of Science indexed · UGent class A1

J2 Hedayati, S., Setak, M., Demir, E., Van Woensel, T. (2024). A new approach to the joint order batching and picker routing problem with alternative locations. *IMA Journal of Management Mathematics*, 35(2), 241–265. Oxford University Press.

J3 Hedayati, S., Setak, M., Van Woensel, T., Demir, E. (2024). Re-supplying autonomous mobile parcel lockers in last-mile distribution. *Future Transportation*, 4(4), 1266–1296. MDPI.

Working papers (unpublished)

W1 Learning decision-dependent recourse domains in two-stage stochastic programming. With S. De Vuyst.

W2 A sample average approximation method with certified scenario reduction for the stochastic inspection station allocation problem with delayed inspection. With S. De Vuyst and B. Raa.

W3 Context-aware inspection station allocation with delayed inspection and sequential loss of inspectability. With S. De Vuyst and B. Raa.

W4 Optimal inspection station allocation in production lines: a dynamic programming approach. With S. De Vuyst and B. Raa.

W5 Multistage stochastic lot-sizing and scheduling for quality-graded disassembly: logic-based Benders decomposition and the value of adaptive scheduling. With S. De Vuyst.

W6 Certificate-conditioned stochastic yield in remanufacturing: integrating weekly shift scheduling with adaptive multi-stage disassembly planning and lot-sizing. With S. De Vuyst.

W7 Integrated disassembly scheduling and workforce planning under condition-based quality classification with misclassification recourse: a two-stage stochastic programming approach.

W8 Uncertainty in the recoverable area: a review of assumptions, models, and solution approaches from returns inventory to recovered outputs.

Conference contributions (peer-reviewed)

C1 Hedayati, S., De Vuyst, S., Raa, B. (2026). Dynamic programming for inspection station allocation in production lines. *ORBEL 40*, pp. 230–232.

C2 Hedayati, S., De Vuyst, S. (2025). Stochastic lot-sizing and scheduling for quality-differentiated disassembly. *IWLS 2025, 15th International Workshop on Lot Sizing*, pp. 73–76.

C3 Hedayati, S., De Vuyst, S. (2025). Dynamic stochastic modeling and optimization of scheduling and quality inspection in re-manufacturing systems. *Flanders Make Scientific Conference*, Leuven.

C4 Hedayati, S., De Vuyst, S. (2025). Stochastic scheduling and quality inspection in remanufacturing systems. *FEARS 2025*, Ghent.
doi:10.5281/zenodo.17435277

C5 Hedayati, S., De Vuyst, S., Raa, B. (2022). Allocation of inspection stations in a multi-product serial assembly line with delayed inspection. *FEARS 2022*, Ghent.
doi:10.5281/zenodo.7406064

C6 Hedayati, S., De Vuyst, S., Raa, B. (2022). Optimal inspection allocation efforts with inspection intervals in flexible multi-stage manufacturing systems. *NOMLC 2022, Netherlands Operations Management and Logistics Conference*.

C7 ZahediSeresht, M., Hedayati, S. (2022). Dynamic capacitated lot sizing and disassembly under yield uncertainty. *CORS/INFORMS International Conference*, Vancouver, 5–8 June 2022.

C8 Hedayati, S., Setak, M., Demir, E., Van Woensel, T. (2021). An application of the clustered generalized vehicle routing problem: order consolidation for e-tailers. *9th INFORMS Transportation Science and Logistics Society Workshop* (virtual), 13–15 December 2021.

Academic service

Peer reviewer for *IMA Journal of Management Mathematics* (2025, reviewer certificate awarded), *Journal of Heuristics* and *OR Spectrum*.

RESEARCH PROJECTS

Oct 2024 – present

REMASC · Flanders Make with Ghent University
Prediction & management of the 'reverse' remanufacturing supply chain: scheduling, quality simulation and predictive analytics for disassembly and remanufacturing.

Jan 2022 – present

Doctoral project · Ghent University
Stochastic optimization models for allocation of quality inspection in flexible assembly.

WHAT I BUILD WITH

Deep — published, benchmarked and defended · Working — used in delivery · Scoped — exactly the stated extent

Optimization models — DEEP

Mixed-integer and integer programming · valid inequalities, lazy constraints and callbacks, warm starts · branch-and-bound · Gurobi (gurobipy), CPLEX, AMPL, HiGHS

Algorithms that scale — DEEP

Lagrangian relaxation with dual-guided repair heuristics · Benders / L-shaped decomposition · exact dynamic programming · simulated annealing, genetic algorithms, differential evolution, large-neighbourhood and local search

Under uncertainty — DEEP

Two-stage stochastic programming · sample average approximation with statistical optimality-gap bounds · scenario generation and reduction · decision-dependent uncertainty

Simulation — DEEP

Discrete-event simulation (SimPy, R simmer) · simulation–optimization · design of experiments

Tools people run — WORKING

Python (pandas, NumPy, SciPy, scikit-learn, multiprocessing, Tkinter) · R (since 2015) · SQL · DuckDB · Git · LaTeX · reproducible experiment pipelines · self-service decision applications with automated scenario comparison, built for non-specialists

Process and quality — WORKING

BPMN and process design · statistical process control, process capability, acceptance sampling · ISO 9001, ISO 14001, OHSAS 18001 Lead Implementer · PMBOK

Learning for OR SCOPED

Reinforcement learning (PPO, custom Gym environments) controlling an optimizer's parameters · graph neural networks for combinatorial optimization (supervised thesis) · nonparametric estimation (nearest neighbours, honest forests) · prescriptive and contextual models

ERP process design SCOPED

SAP Materials Management process blueprinting on one six-month implementation; process design, not configuration

PUBLIC LABS AND OPEN TOOLS

RIPPLE Optimization Lab
Stochastic MILP demonstrator for recovery planning under uncertain yield and demand (HiGHS, CVaR option).

Retail Analytics Lab
Auditable pipeline over 1,067,371 transaction lines; Python, SQL/DuckDB and R with output parity and 60 passing tests.

ISAPDI Simulation Lab
Interactive inspection-allocation and production-line simulation.

RevMode for LaTeX
Open-source, MIT-licensed single-source manuscript revision workflow.

LANGUAGES AND CERTIFICATION

Persian — native

English — full professional proficiency

Dutch — B1+, certified by the University Language Centre (UCT), Ghent University

ISO 9001 Foundation — Quality

ISO 14001 — Environmental Management Systems

OHSAS 18001 Lead Implementer — Health & Safety

PMBOK · Microsoft Project

REFERENCES

Available on request — including both Ph.D. supervisors at Ghent University and co-authors at Eindhoven University of Technology and Cardiff University.